Fair results from a bag

Mathematics before the session

For independent draws from the same unchanged bag, write p for the chance of red. If $0 < p < 1$, then $P(RB) = p(1-p) = P(BR)$. Assign a square to BR, a circle to RB, and no shape to RR or BB. Conditional on producing a shape, each shape has chance 1/2. Interchanging the two shape names also works. This is the von Neumann extractor.

For three red counters and one blue, the 16 equally likely **labeled** ordered pairs fall into color classes of sizes 9, 3, 3, 1 for RR, RB, BR, BB. A deterministic rule that sees only the color pair cannot split a class. The two rules above are the only fair rules that sometimes produce each shape: the weight 9 must be discarded, and among 3, 3, 1 the only equal nonempty disjoint sums are 3 and 3. Thus ten of the sixteen labeled outcomes must be skipped.

The output rate is $2p(1-p)$. Among four-counter bags it is largest for two red and two blue: 8 of 16 pairs produce a shape, versus 6 of 16 for a 3-to-1 bag. A one-color bag produces none; equal zero probabilities are not a functioning fair chooser.

Assumptions and limits

Uniform counter identities, replacement, mixing, and independent choices model the advertised random mechanism. Deliberately selected stories are possibility arguments, not random samples. If the first and second draws have red probabilities p and q and are independent, the kept weights are $p(1-q)$ and $(1-p)q$: they agree exactly when $p = q$, with both colors needed for positive output. Independent draws from the same bag are sufficient but not necessary: the rule needs $P(RB) = P(BR)$. The no-replacement variation is fair because reversing distinct-counter pairs preserves their chances.

No finite number of pairs guarantees a shape or balanced totals. With independent fixed-bag pairs and $0 < p < 1$, the chance of no output after m pairs is $[p^2+(1-p)^2]^m$, tending to zero. Eventual output has probability one, but there is no finite worst-case deadline. A six-pair experiment can suggest a rule; equal-weight pairing proves it.

What each group is discovering

K-1: ordered pairs, a color-only rule, matching equal-weight pictures, and possible versus forced outcomes. **Grades 2-3:** complete labeled outcomes, reversal symmetry, efficiency, and a changing-bag counterexample. **Grades 4-5:** assumptions, optimality within the color-only rules, and symmetry without independence.

Running the session

Materials and readiness

For eleven children at three adult-led tables: per pair of children, two opaque bags or cups and about ten red and eight blue same-size, same-feel counters (both upper Problem 5s use two bags at once); pencils; and reusable two-position drawing mats. Mark three reds R1, R2, R3 for exact checking; K-1 page 4 pictures the other bags' marked pairs. Keep extra counters outside the draw bag. Paper squares and circles can record outputs; a cross means no output. Use R/B letters as well as color. No special purchases or cutting of the student pages is required.

Print K-1 pages 1-6, grades 2-3 pages 1-4, and grades 4-5 pages 1-4; give out one or two pages at a time. K-1 needs order, color matching, and adult reading. The middle group needs a row/column grid and small counts. The upper group needs comparison of equal-size outcome sets; fractions help but are not required. This packet is prepared and unpiloted.

Short shared launch

Let children handle the counters. Put three reds and one blue in a bag. Draw R, return it, mix, draw B, and put the record in left-to-right order RB. Contrast BR. Then draw B, return it and draw B again: BB is possible because the counter goes back. Say: "A machine sees these two colors in order. It gives a square, a circle, or nothing. Can you make the shapes equally likely?" Show only the printed sample BR-to-square; do not supply the whole rule. Demonstrate choosing a pair on purpose separately from a hidden random draw.

A flexible hour

0-5 minutes: running around. **5-10:** shared handling and launch. **10-22:** invent rules and make a few pairs; use six random pairs only as exploration. **22-33:** move to the complete marked-pair pictures or grid and compare shape weights. **33-35:** movement break: children hold "first" and "second" roles and reverse an RB pair. **35-47:** select changed bags or a stopping question. **47-53:** each table shows the group a reason that works beyond its trials. **53-56:** reset and collect materials.

A good stopping point is one justified fair rule and one counterexample to "six pairs must balance." The youngest table can spend most of the hour on K-1 Problems 1-3; do not rush the complete-picture comparison. Keep the changing-bag and no-replacement questions for children ready to track a modified mechanism.

Hints to offer only when needed

"Can two marked reds be different outcomes with the same colors?" "What happens when you reverse a mixed pair?" "Must nine RR pictures all get the same answer?" Use a partner to build the reverse of each kept pair. For a no-output counterexample, ask which pair the rule discards rather than asking for more trials.

Source and adaptation

David A. Levin, Yuval Peres, Elizabeth L. Wilmer, *Markov Chains and Mixing Times*, Appendix B.2, "Von Neumann Unbiasing," printed p. 312 (PDF p. 328). University-hosted text. The two-counter classroom rule, finite color-only optimality check, bag comparisons, and keyed examples here are independently derived adaptations.

K to 1 answer key

Keyed to the final six-page K-1 packet. Throughout this key, use BR → square, RB → circle, RR and BB → no shape. Swap the shape names throughout if children chose the other fair rule.

Problem 1

All four color pairs are RR, RB, BR, BB. Replacement makes BB possible even with only one blue counter. Taking the blue counter out leaves only RR: RB, BR, and BB disappear. A child may choose these draws deliberately to show possibility.

Problem 2

Use the rule stated above or its shape-swapped version. The six random outputs vary; even no outputs is possible. Do not judge the rule from a balanced or unbalanced sample. If a child proposes a different rule, keep it as a conjecture until the picture check in Problem 3.

Problem 3

RR has nine marked pictures; RB and BR have three each; BB has one. The two fair rules are exactly: (i) BR square, RB circle, RR/BB no shape; (ii) BR circle, RB square, RR/BB no shape. To see completeness, a shape receiving all nine RR pictures cannot be balanced by the seven other pictures, so RR is skipped. Among the remaining classes, BB cannot be balanced without breaking a three-picture class. Thus BB is skipped and RB/BR must go to different shapes. Adult-supported matching is enough; a written exhaustion is not required.

Problem 4

One red/three blue gives each shape in 3 of 16 marked pairs (the R, B1, B2, B3 pictures on page 4). Two red/two blue gives each shape in 4 of 16 (the R1, R2, B1, B2 pictures). In both bags the shapes have equal positive chances. Four red or four blue gives no shape at all, so neither can give each shape.

Problem 5

Choose two red and two blue: the R1, R2, B1, B2 pictures on page 4 give a shape in 8 of 16. For red counts 0, 1, 2, 3, 4, the numbers of mixed labeled pairs are 0, 6, 8, 6, 0 out of 16. Hence 2-and-2 is optimal. No four-counter bag gives a shape on every pair under this rule: repeating the identity of the first counter is always possible and makes a same-color pair. A changed rule could: with two red and two blue, first color red gives a square and blue a circle is fair and never skips (8 and 8 of 16). That is why the problem says to keep your rule.

Problem 6

Examples of six-pair stories: **no shapes:** RR, RR, RR, RR, RR, RR. **Exactly one square, no circles:** BR, RR, RR, RR, RR, RR. **Exactly three circles, no squares:** RB, RB, RB, RR, RR, RR. Any ordering with the required outputs works. Six random pairs do not always give a shape: the first story has positive probability.

Grades 2 to 3 answer key

Problems 1 and 2

Invent BR → square, RB → circle, RR/BB → no shape, or swap the shapes. Six trials cannot settle fairness. With labeled counters R1, R2, R3, B, all 16 ordered pairs are equally likely. In the grid below, rows are first draws and columns second draws.

First / second	R1	R2	R3	B
R1	RR	RR	RR	RB
R2	RR	RR	RR	RB
R3	RR	RR	RR	RB
B	BR	BR	BR	BB

The three RB cells and three BR cells have equal total weight. Pair (Ri,B) with (B,Ri). There are three outcomes for either shape and ten skipped outcomes; among the six kept outcomes each shape occupies half.

Problem 3

Both bags work. Grid heads: R1, R2, B1, B2 and R, B1, B2, B3. Two red/two blue gives four labeled RB and four BR outcomes out of 16. One red/three blue gives three RB and three BR outcomes. Reversing a pair pairs the outputs exactly, regardless of the observed random run.

Problem 4

The two-red bag produces a shape in 8 of 16 pairs; the three-red bag in 6 of 16. Choose the two-red bag. Both bags are fair when a shape comes out.

Problem 5

No. Grid rows R1, R2, R3, B; columns R, B1, B2, B3. Use the printed first bag **three red, one blue** and second bag **one red, three blue**. The exact counts are RR 3, RB 9, BR 1, BB 3 among 16 equally likely cross-bag identity pairs. For the key rule, circle has 9 ways and square 1; conditional on a shape, their chances are 9/10 and 1/10. Reversal no longer gives an equally likely outcome.

Problem 6

No. RR repeated six times is a legal no-shape story. With replacement and the original bag its chance is $(9/16)^6$, which is positive. Children need only exhibit the legal story, not compute its chance. Hint: ask for a six-pair story that settles it.

Problem 7

No color-only deterministic rule can be both fair and always produce a shape. All nine RR labeled pairs receive the same shape, already more than half of the 16 outcomes. The other shape cannot catch up even if it receives all seven remaining outcomes.

Grades 4 to 5 answer key

Problems 1 and 2

Use BR → square, RB → circle, RR/BB → no shape, or interchange the shapes. The four color classes have labeled sizes 9, 3, 3, 1. Reversal pairs each (Ri,B) with (B,Ri), giving equal shape probabilities 3/16 and conditional probabilities 1/2. Other rules fail Problem 2’s check: RR → square, the rest → circle gives 9 against 7.

Problem 3

Both bags work: five red/two blue has 10 RB and 10 BR outcomes among 49; three red/four blue has 12 and 12 among 49. In general a fixed bag with r red and b blue has rb outcomes of each mixed order among $(r+b)^2$ equally likely identity pairs. The bag contents need not be known, but both r and b must be positive and the draws must follow the stated mechanism.

Problem 4

Always giving a shape is impossible: the RR class alone has weight $9 > 8$. If skipping is allowed, RR must still be skipped because all other classes total only 7. The remaining weights are 3, 3, 1. Their only equal, nonempty, disjoint subset sums are the two 3s. Therefore BB must also be skipped. The minimum is 10 skipped labeled pairs, attained by the two usual rules. This claim is restricted to deterministic color-only rules, not rules using identity marks or extra randomness.

Problem 5

First case: first bag 3R/1B, second 1R/3B gives RB 9/16 and BR 1/16, so unfair; a different rule, RR → square and BB → circle, is fair there, 3/16 each. **Second:** 3R/1B then 6R/2B gives RB and BR each 6/32 = 3/16, so fair; equal color proportions matter, not equal bag size. **Third:** 2R/2B then 1R/1B gives each mixed order 2/8 = 1/4, so fair. In general the equality $p(1-q) = (1-p)q$ reduces to $p = q$; positive output additionally requires $0 < p < 1$.

Problem 6

Yes. Without replacement there are $4 \times 3 = 12$ equally likely ordered pairs of distinct identities. RR has six, RB three, BR three, BB none. Reverse each kept pair to match its weight. Each shape has probability 3/12, and conditional on output it has chance 1/2. The within-pair draws are dependent, so the independent-draw product proof is inapplicable; the reversal proof still works. Another fair rule never skips here: both red → square, otherwise circle, 6 of 12 each.

Problem 7

Neither claim follows. Six RR pairs produce no shape; BR, RR, RR, RR, RR, RR produces one square and no circles. Each story has positive probability under the original mechanism. Fairness says the possible outcomes have equal chances when a shape is produced; it does not force equal realized totals or a finite waiting bound.

Facilitator checkpoint: keep three claims separate - a legal example shows possibility, a finite trial suggests a conjecture, and equal-weight exhaustive counting or a bijection proves the exact probability claim.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

First routes: keep K-1 P1-3, as on guide p2; grades 2-3 use P1-2 before comparing bags in P3-4; grades 4-5 use P1-2 before generalizing the rule in P3. On return, K-1 can change bags or build no-output stories (P4-6). Grades 2-3 can choose efficiency (P4), changing bags (P5), or stopping/skip claims (P6-7). Grades 4-5 can separate skip optimality (P4), unequal draw mechanisms (P5), and no-replacement/stopping questions (P6-7). Keep the ordered-pair record and the chosen/random distinction visible.